

Jaemin Sung

Jeonju, South Korea | (+82) 010-3006-4611 | jac.sung1997@gmail.com

GitHub: <https://github.com/jsung1997> | LinkedIn: www.linkedin.com/in/jay-sung-032328161

EDUCATION

Jeonbuk National University

Master of Science in AI & Software Engineering

Jeonju, South Korea

On Hold: March. 2025 - Nov. 2025

Gwangju University

Bachelor of Science in Computer Science

Gwangju, South Korea

Graduated: Dec. 2024

PROFESSIONAL EXPERIENCE

JBNU's Physical AI Convergence Technology

Research Assistant

Jeonju, South Korea

March. 2025 - Nov. 2025

- Configured and executed model training environments using PyTorch and TensorFlow, monitored performance metrics (accuracy, loss, F1-score), and conducted cross-validation and error analysis for continuous improvement.
- Analyzed model outputs using confusion matrices, feature importance, and hyperparameter tuning to derive performance improvement strategies and actionable insights for AI-driven factory operations.

RXO.WORLD

Internship

Gwangju, South Korea

Dec. 2023 - Mar. 2024

- Monitored and tested 100+ AI-enabled IoT devices, identifying performance issues and adjusting configurations to optimize operations, improving overall productivity by 15%.
- Spearheaded meetings with overseas partners, streamlining cross-cultural communication and driving productive collaboration between international teams.

PROJECTS & ACHIEVEMENTS

AI-Driven Reliability Measurement Platform for Nuclear PLC Systems

Lead Cloud AI Engineer

Jeonju, South Korea

Aug. 2025 - Nov. 2025

- Built a web-based system for reliability assessment of PLC software in nuclear power plants, integrating statistical defect analysis.
- Developed and deployed an AWS Lambda-based PLC reliability analysis system using Bayesian methods for failure prediction and automated S3 result delivery via preassigned API links.

DRV Framework (Defect-Risk Visualization) for Software Reliability Analysis

Machine Learning Engineer & System Architect

Jeonju, South Korea

May. 2025 - Nov. 2025

- Developed a defect-risk prediction model leveraging XGBoost and feature attribution methods (SHAP/LIME) to identify high-risk software modules in large-scale industrial systems.
- Designed a visualization layer (React + Plotly) to enable domain engineers to monitor defect risk and reliability metrics in real time.

WARDEN (Capstone Design Team)

Full Stack Developer with Data Administration Focus

Gwangju, South Korea

Sep. 2023 - Dec. 2023

- Awarded 1st place, LINC 3.0 Academic Competition
- Awarded 1st place, Korea Electric Power Corporation (KEPCO) Competition
- Led a collaborative effort with team members to address a real-world issue by utilizing AI models and IoT devices, resulting in the successful development of an innovative solution.

SKILLS

Machine Learning: Multi-Agents, Attribution-Guided Editing, Software Defect Prediction, Commit Diff Analysis

Cloud & DevOps: AWS, Git, GitHub

Programming Languages: Python, C++, JavaScript, Java

NLP & LLM: Fine-tuning, LoRA Adaptation, Prompt Engineering, Cross-Model Analysis

Data Management: R, SQL, MySQL/Oracle, TensorFlow/PyTorch, Pandas/Numpy

Language: English (Native), Korean (Fluent)